

VISTA Talk

Direct All-Atom Nonadiabatic Simulation of Absorption Spectroscopy in Liquid Phase

Zengkui Liu

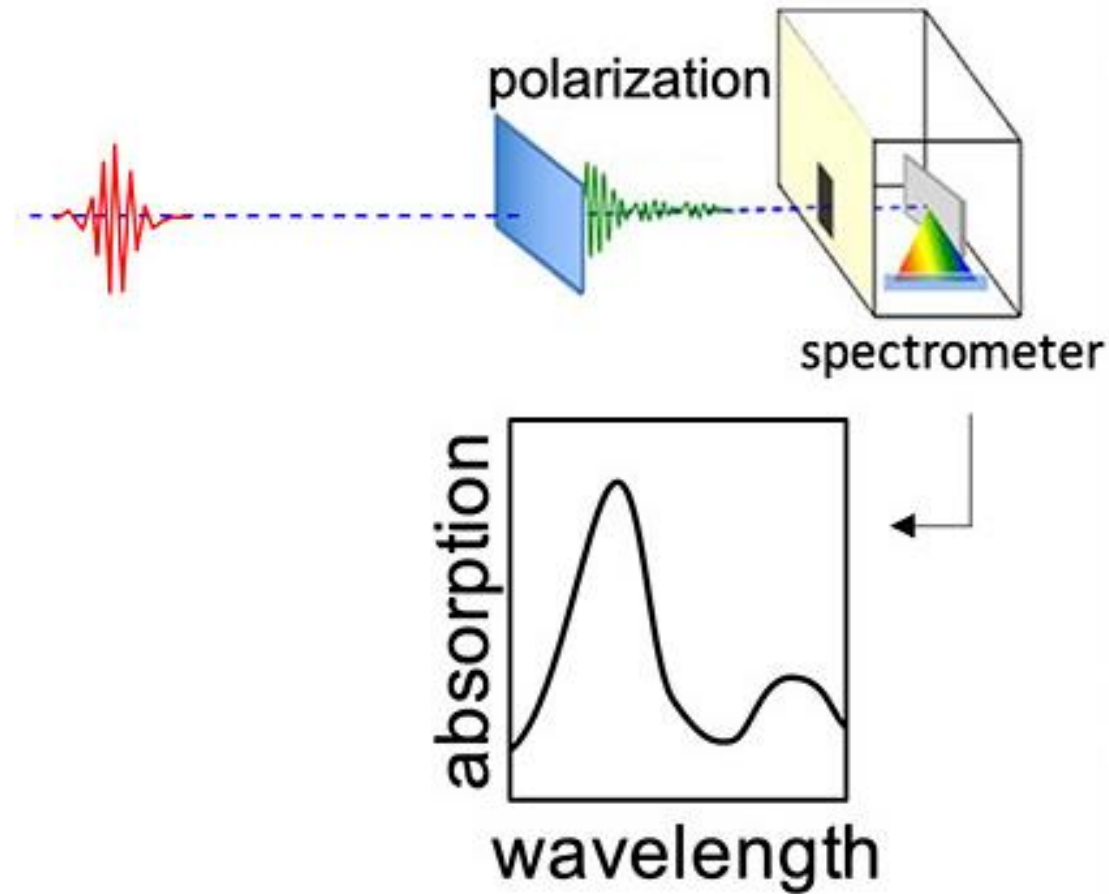
Oct. 15, 2025

Prof. Xiang Sun Group

NYU Shanghai & New York University

Simulating linear absorption spectra experiment

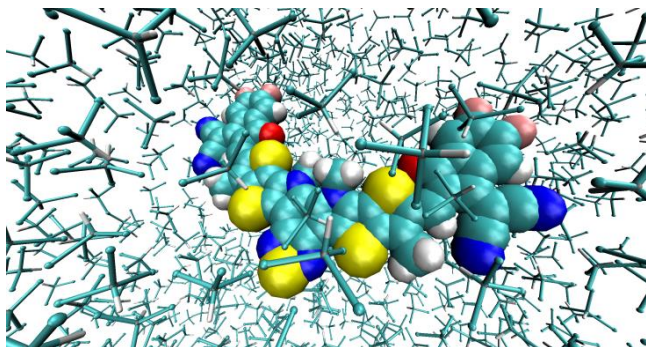
- How to simulate linear absorption spectra measurement?



Solve chemical dynamics problem

Hamiltonian of the condensed phase

- Anharmonic potential energy surface (PES)
- Many degrees of freedoms



Hamiltonian

Dynamics

Observables

Electronic spectroscopy

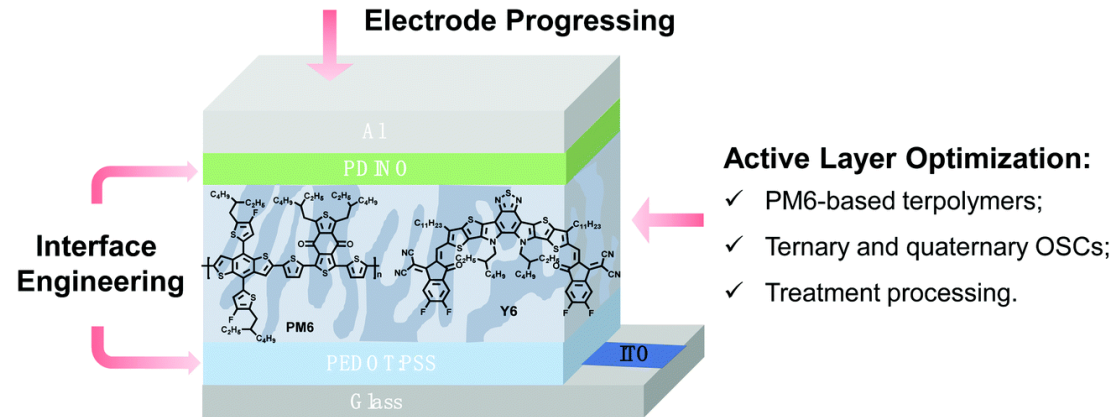
Time evolution of the system:
time-dependent
Schrodinger equation and its approximations

Why Approximation?
For exact dynamics cost
Too much....

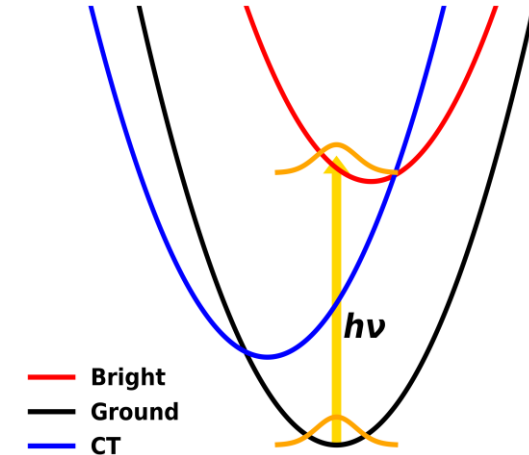
N	Memory
5	1.6 MB
10	160 GB
50	10^{36} PB
Scaling	$\sim O(10^N)$

Y6 photoinduced charge transfer

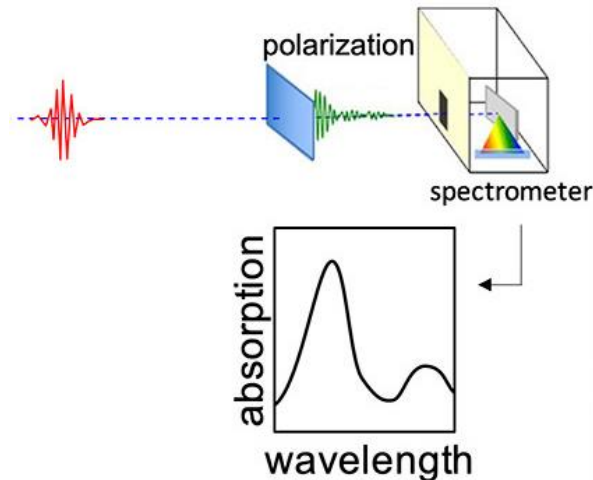
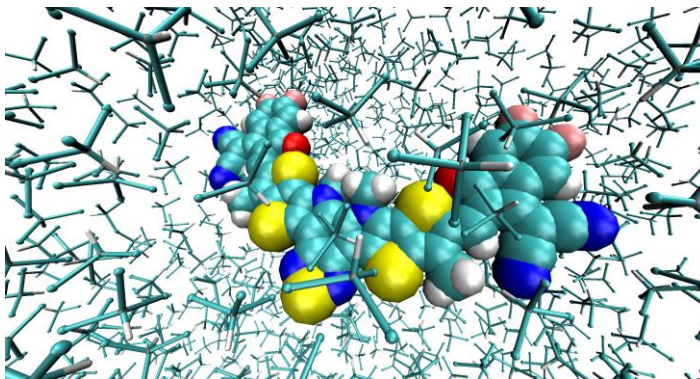
Y6 in organic photovoltaic (OPV) in battery device
Most efficient binary OPV battery ~19.3%



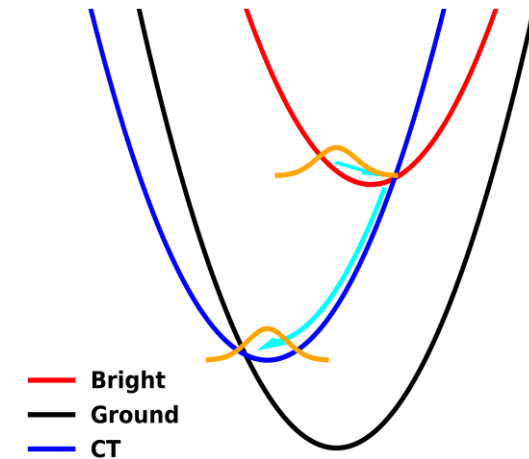
• Photoinduced excitation



Y6 in chloroform: linear absorption spectra



• Charge transfer process



- Ref. 1. J. Fu, et.al. Nat. Comm., 14, 2760 (2023)
2. Q. Guo, et.al. Mater. Chem. Front., 5, 3257 (2021)
3. I. Conti, et.al. J. Am. Chem. Soc., 142, 16117 (2020)

Meyer-Miller Stock-Thoss Mapping Hamiltonian

- Multistate Hamiltonian

$$\hat{H} = \frac{\hat{\mathbf{P}}^2}{2} + \sum_{j,k}^F V_{jk}(\hat{\mathbf{R}}) |j\rangle\langle k| = \frac{\hat{\mathbf{P}}^2}{2} + \hat{V}(\hat{\mathbf{R}}).$$

- Mapping F electronic states to F singly excited harmonic oscillators

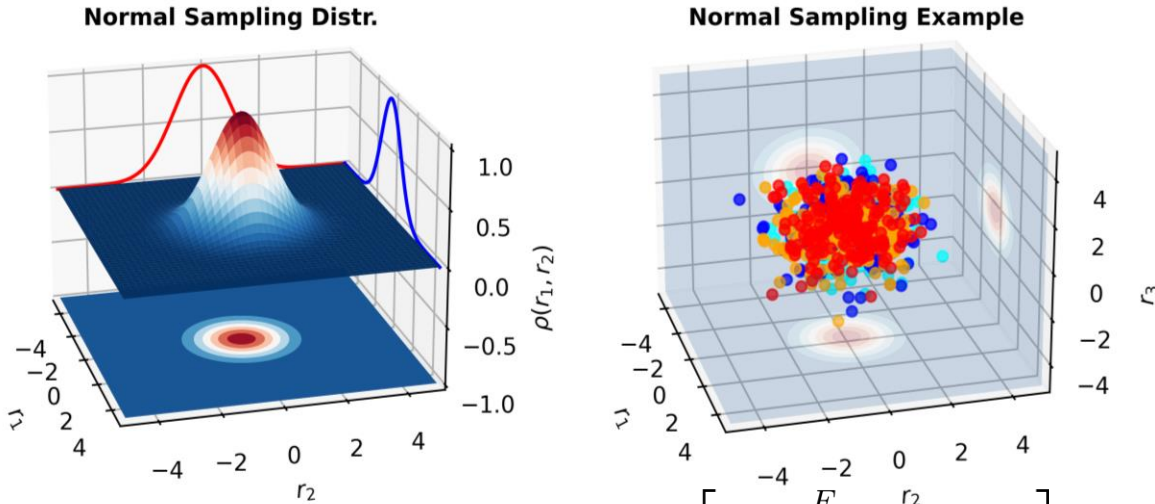
$$|j\rangle \longmapsto \underbrace{|0_1 \cdots 1_j \cdots 0_F\rangle}_{F\text{-states}} \equiv |1_j\rangle$$
$$|j\rangle\langle j| \longmapsto \hat{a}_j^\dagger \hat{a}_j = \frac{1}{2\hbar} (\hat{q}_j^2 + \hat{p}_j^2) - \frac{1}{2},$$
$$|k\rangle\langle j| \longmapsto \hat{a}_k^\dagger \hat{a}_j = \frac{1}{2\hbar} (\hat{q}_k - i\hat{p}_k) (\hat{q}_j + i\hat{p}_j).$$

- Meyer-Miller Stock-Thoss Hamiltonian

$$H(\mathbf{R}, \mathbf{P}, \mathbf{q}, \mathbf{p}) = \frac{\mathbf{P}^2}{2} + \sum_j^F V_j(\mathbf{R}) \left(\frac{1}{2\hbar} q_j^2 + \frac{1}{2\hbar} p_j^2 - \gamma \right) + \sum_{j \neq k}^F V_{jk}(\mathbf{R}) \frac{1}{2\hbar} (q_j - ip_j) (q_k + ip_k).$$

Mapping Kernel and Mapping Variable Sampling

Linearized semiclassical (LSC)



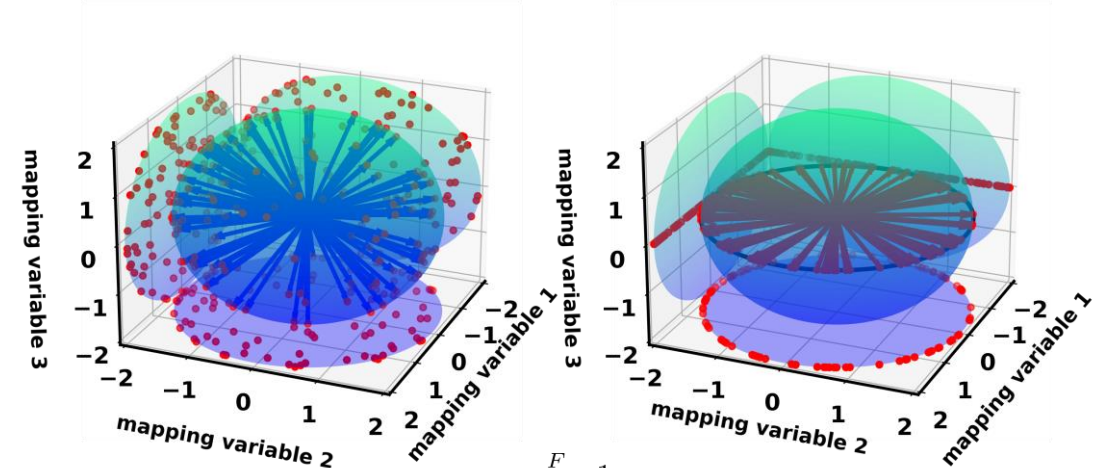
Sampling: $G(\mathbf{q}, \mathbf{p}) = 2^{F+2} \exp \left[-\frac{1}{\hbar} \sum_l^F (q_l^2 + p_l^2) \right]$

Kernel: $M_{jk}^{(1)}(\mathbf{q}, \mathbf{p}) = \frac{1}{2\hbar} (q_j - ip_j)(q_k + ip_k) - \gamma \delta_{jk},$
 $M_{jk}^{(2)}(\mathbf{q}, \mathbf{p}) = \left[\frac{1}{2\hbar} (q_j - ip_j)(q_k + ip_k) - \frac{\gamma}{2} \delta_{jk} \right] G(\mathbf{q}, \mathbf{p})$

Classical mapping model (CMM) / Spin-mapping (SPM)

spherical sampling

focused sampling



Sampling: $S_{\text{spherical}}(\gamma) : \sum_{j=1}^F \frac{1}{2\hbar} (q_j^2 + p_j^2) = 1 + F\gamma$
 $S_{\text{foc}} : \sum_{j=1}^F \left(\frac{1}{2\hbar} (q_j^2 + p_j^2) - \gamma \right) = \delta_{j,\text{occ}}$

Kernel: $M_{jk}(0) = \frac{1}{2\hbar} (q_j - ip_j)(q_k + ip_k) - \gamma \delta_{jk}$
 $M_{jk}(t) = \frac{\bar{Q}(\gamma, F)}{2\hbar} (q_j - ip_j)(q_k + ip_k) - \bar{\gamma} \delta_{jk}$

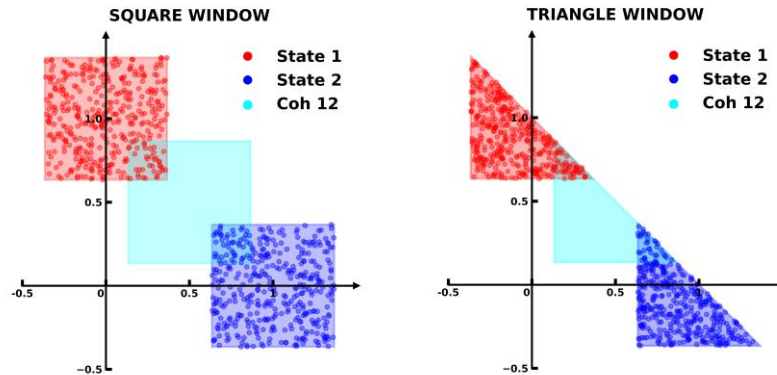
- Ref: 1. H. Wang et. al., J. Chem. Phys., 108, 9726 (1998).
 2. Q. Shi et. al. J. Phys. Chem. A, 108, 6109 (2004)
 3. X. Gao et. al. J. Chem. Theory Comput., 16, 2283 (2020)

4. J. Liu, J. Chem. Phys., 145, 204105 (2016)
 5. X. He et al. J. Chem. Phys., 151, 024105 (2019)
 6. R. Runeson, et. al. J. Chem. Phys., 151, 044119 (2019)

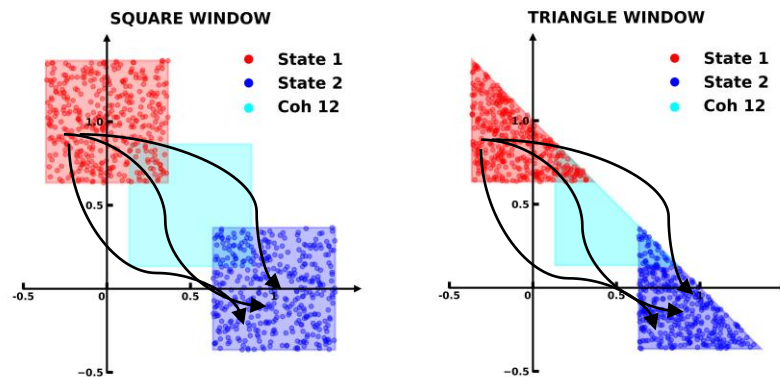
Quasiclassical / Mixed Quantum-Classical

Symmetrical Quasiclassical (SQC)

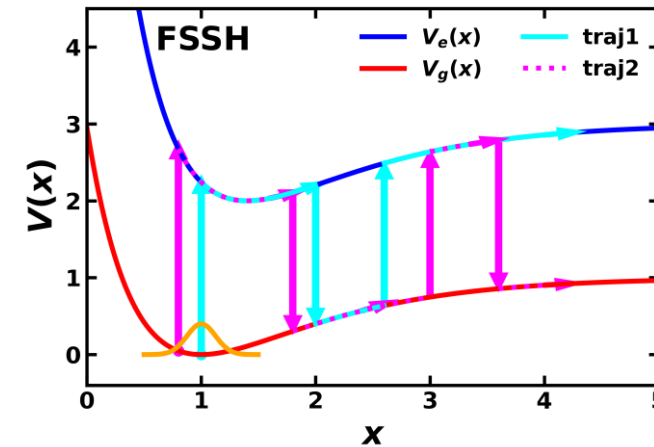
Sampling



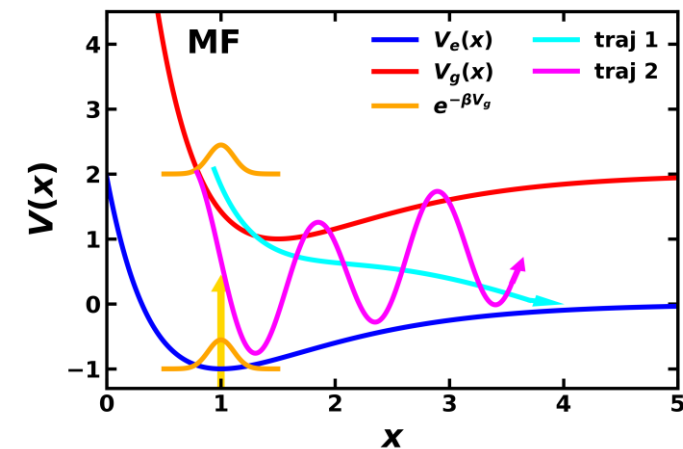
Dynamics



Fewest-switches Surface Hopping (FSSH)



Ehrenfest Mean-field (MF)



Ref: 1. S.J. Cotton, W.H. Miller, J. Chem. Phys., 139, 234112 (2013)
2. S.J. Cotton, W.H. Miller, J. Chem. Phys., 145, 144108 (2016)

3. J. Tully, J. Chem. Phys., 93, 1961 (1990)
4. P. Ehrenfest, Z. Phys. 45, 455, (1927)

Observables of the condensed phase system

- Wave function of electronic and nuclear degrees of freedom (DOF)

$$|\psi\rangle = |\psi_e\rangle \otimes |\psi_N\rangle$$

- Density matrix of a mixed state

$$\hat{\rho} = \sum_i p_i |\Psi_i\rangle \langle \Psi_i|$$

- Reduced density matrix (RDM)

$$\hat{\sigma} = \text{Tr}_N[\hat{\rho}] = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1F} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2F} \\ \cdots & & & \\ \sigma_{F1} & \sigma_{F2} & \cdots & \sigma_{FF} \end{bmatrix}$$

Coherence

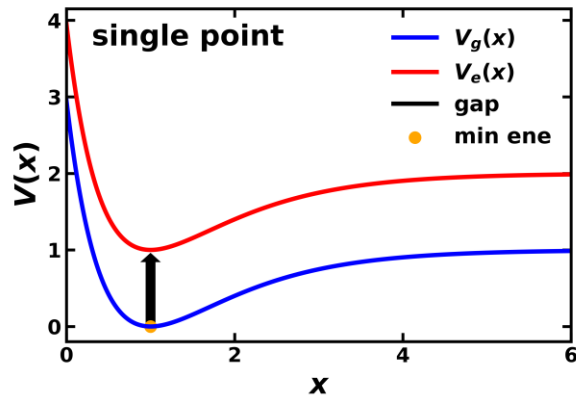
Population

- Quantum time correlation function (TCF):

$$\sigma_{mn}(t) = \sum_{j,k} \text{Tr}_{e,N} \left[\hat{\rho}_N(0) \otimes |j\rangle \langle k| e^{-i\hat{H}t/\hbar} |n\rangle \langle m| e^{i\hat{H}t/\hbar} \right]$$

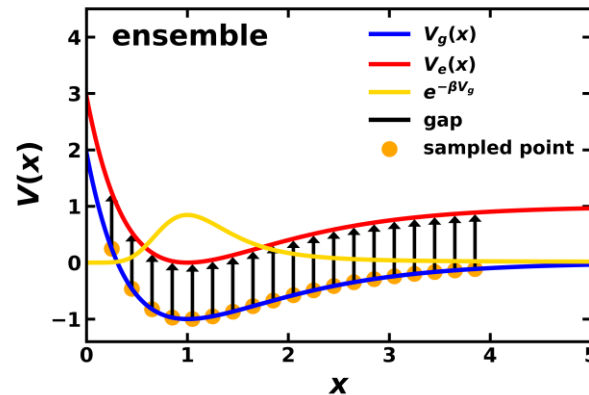
Other observable: electronic spectroscopy, etc.

Traditional Theoretical Spectroscopy Methods



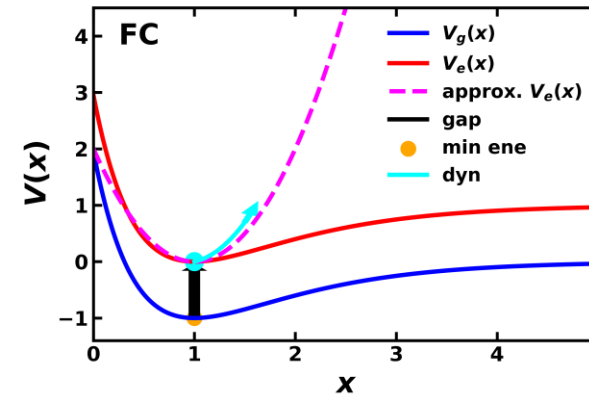
Single Point Calculation

1. No sampling
2. No dynamics
3. Ad hoc widths



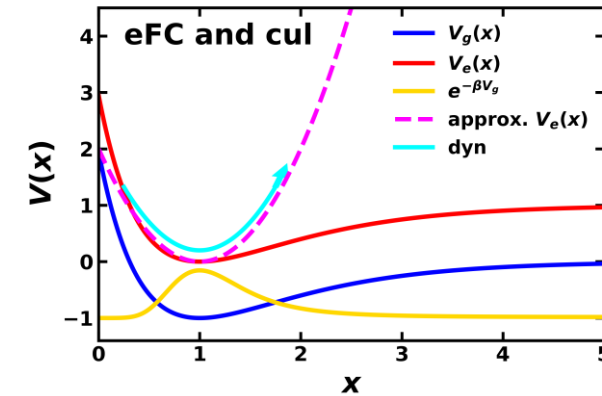
Ensemble Approach

1. Canonical sampling
2. No dynamics
3. Ad hoc widths (binning)



Frank-Condon Approach

1. No sampling
2. Harmonic dynamics
3. Dynamical width

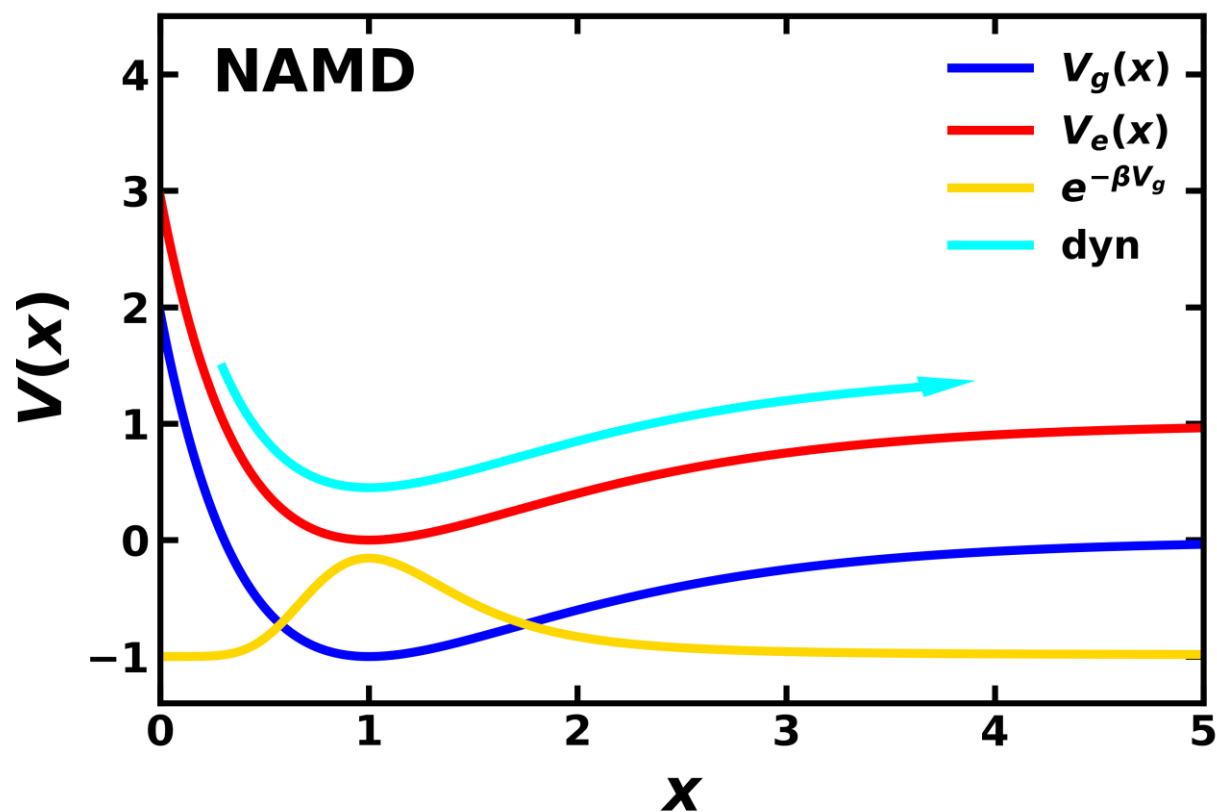


Second-order Cumulant Approach

1. Canonical sampling
2. Harmonic dynamics
3. Dynamical width

Ref.: 1. R. Cammi et al. J. Chem. Phys., 122, 104514 (2005)
 2. T.J. Zushsdorff et al. J. Chem. Phys., 151, 074111 (2019)
 3. S. Mukamel et al. Chem. Rev., 104, 2073 (2004)
 4. S. Mukamel et al. Principle of Nonlinear Spectroscopy, NY (1995)

All-atom Nonadiabatic Mapping Methods



Features of our method:

1. Canonical sampling
2. All-atom nonadiabatic dynamics with explicit solvents
3. Consistent lineshape broadening
4. Direct simulation of photoinduced and relaxation dynamics (NP)
5. Well-defined coherences

Linear absorption spectra: perturbative theory

- Transition dipole

$$\hat{\boldsymbol{\mu}} = \sum_{e \neq g}^F (\boldsymbol{\mu}_{ge} |g\rangle \langle e| + \boldsymbol{\mu}_{eg} |e\rangle \langle g|)$$

- Optical response function

$$J(t) = \frac{1}{3} \text{Tr} [\hat{\rho}(0) \hat{\boldsymbol{\mu}}(0) \hat{\boldsymbol{\mu}}(t)]$$

$$R(t) = \frac{i}{\hbar} \text{Tr} [\hat{\rho}_{\text{eq}} \hat{\boldsymbol{\mu}}(0) \hat{\boldsymbol{\mu}}(t)] = -\frac{2}{\hbar} \theta(t) \text{Im}[J(t)]$$

- Linear absorption spectra

$$I(\omega) = \text{Im} \int_0^\infty dt e^{i\omega t} R(t)$$

Linear absorption spectra: nonperturbative theory

- Semiclassical light-matter interaction Hamiltonian

$$\hat{H}(t) = \hat{H}_M + \hat{H}_{\text{int}}(t)$$

$$\hat{H}_M = \sum_{i=1}^N \frac{\hat{P}_i^2}{2m_i} + \sum_{j,k}^F V_{jk}(\hat{\mathbf{R}}) |j\rangle \langle k|$$

$$\hat{H}_{\text{int}}(t) = -\hat{\boldsymbol{\mu}} \cdot \mathbf{E}(t)$$

- MMST light-matter interaction Hamiltonian

$$H(\mathbf{R}, \mathbf{P}, \mathbf{q}, \mathbf{p}) = \sum_{i=1}^N \frac{P_i^2}{2m_i} + \sum_{j,k}^F V_{jk}(\mathbf{R}) \left[\frac{1}{2\hbar} (q_j - ip_j)(q_k + ip_k) - \gamma \delta_{jk} \right] - \sum_{j \neq k}^F \boldsymbol{\mu}_{jk} \cdot \mathbf{E}(t) \frac{1}{2\hbar} (q_j - ip_j)(q_k + ip_k),$$

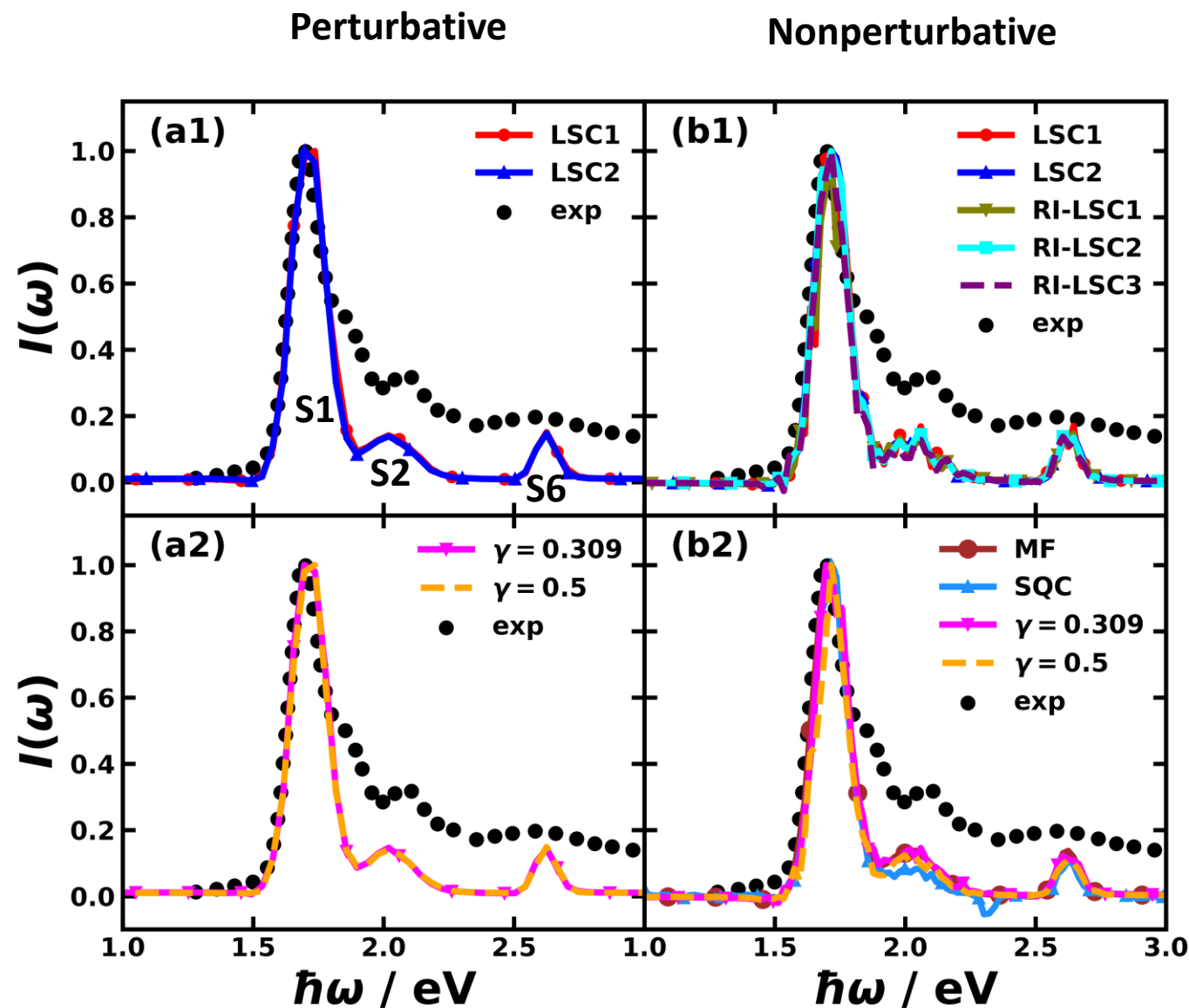
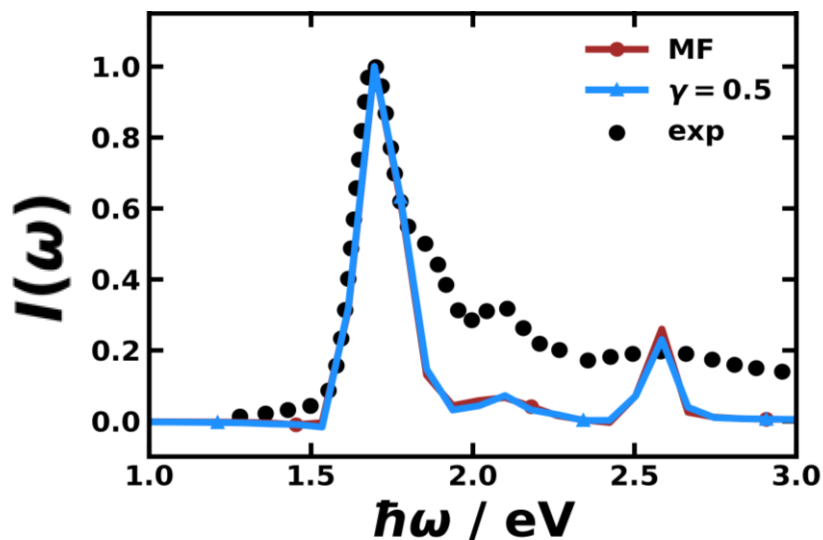
- Nonequilibrium polarization

$$P(t) = \text{Tr} [\hat{\mathbf{e}} \cdot \hat{\boldsymbol{\mu}} \hat{\sigma}(t)]$$

- Nonperturbative absorption $I(\omega) = \text{Im} \int_0^\infty dt e^{i\omega t} P(t)$

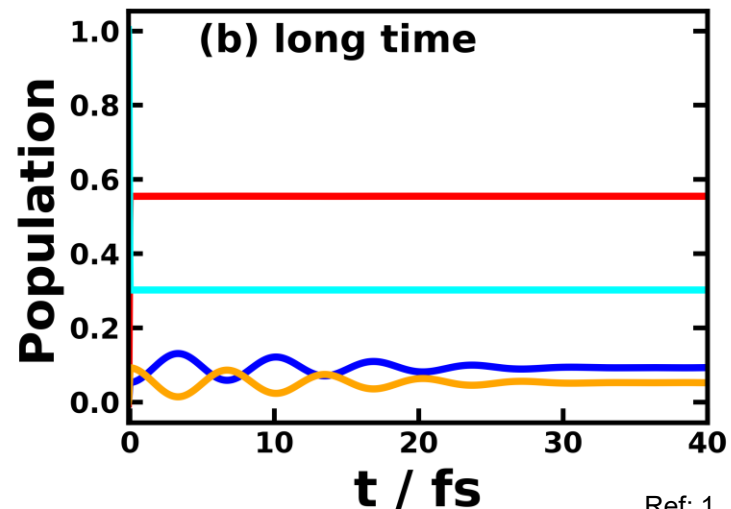
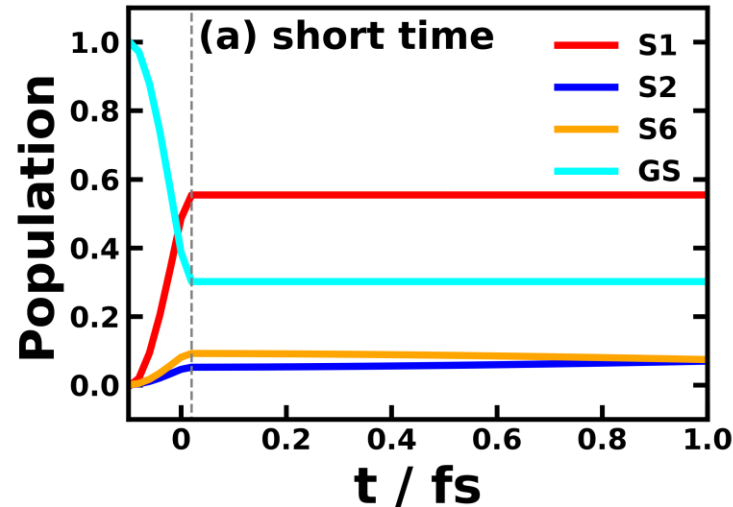
All-atom Linear Absorption Spectra

- Simulated spectra using all-atom nonadiabatic dynamics reproduced the experimental spectral peaks at 1.7, 2.1, and 2.5 eV
- Nonadiabatic dynamics matters in Y6 PICT process.



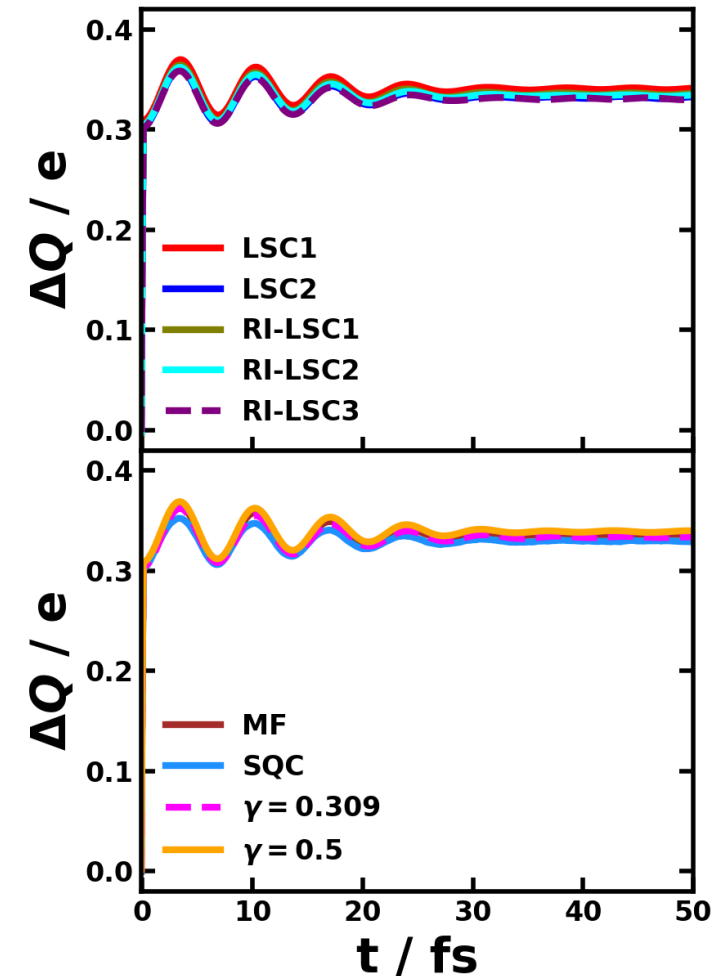
All-atom information from direct simulation

- Population dynamics
photoexcitation process simulation



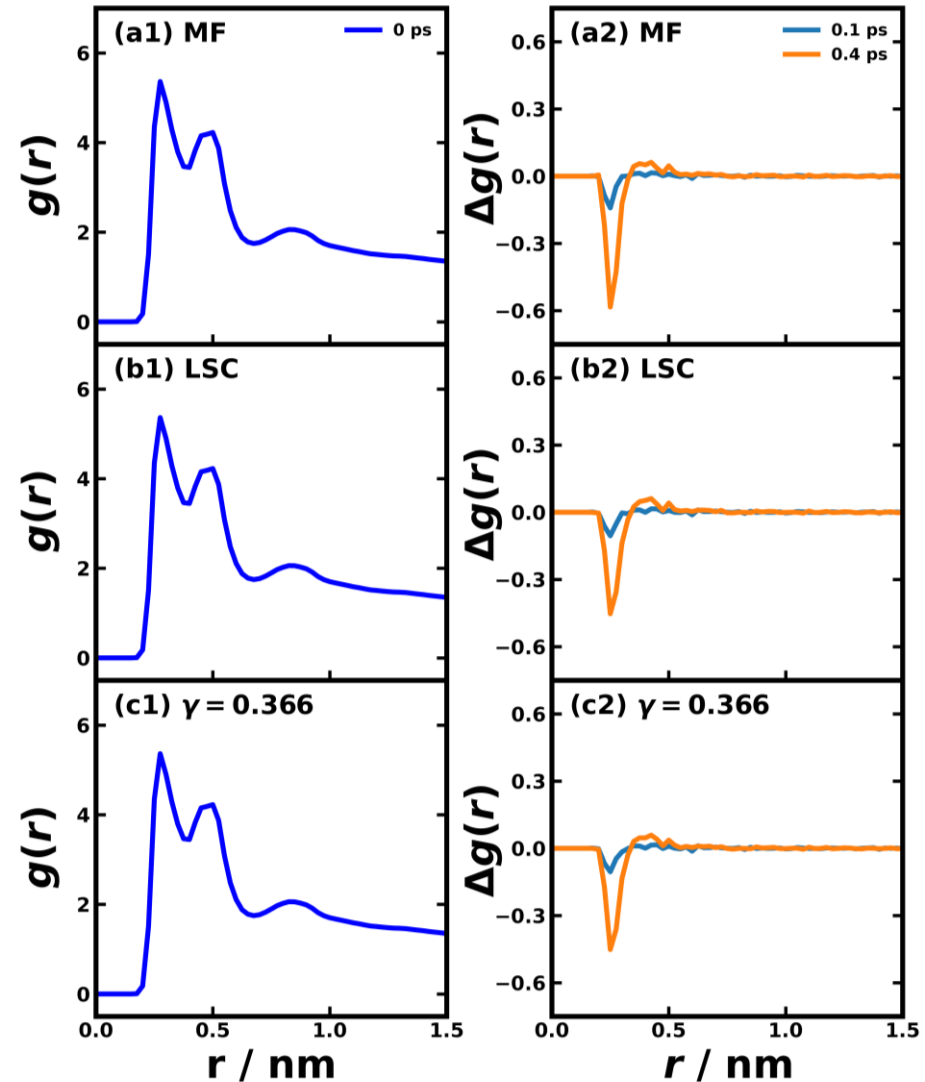
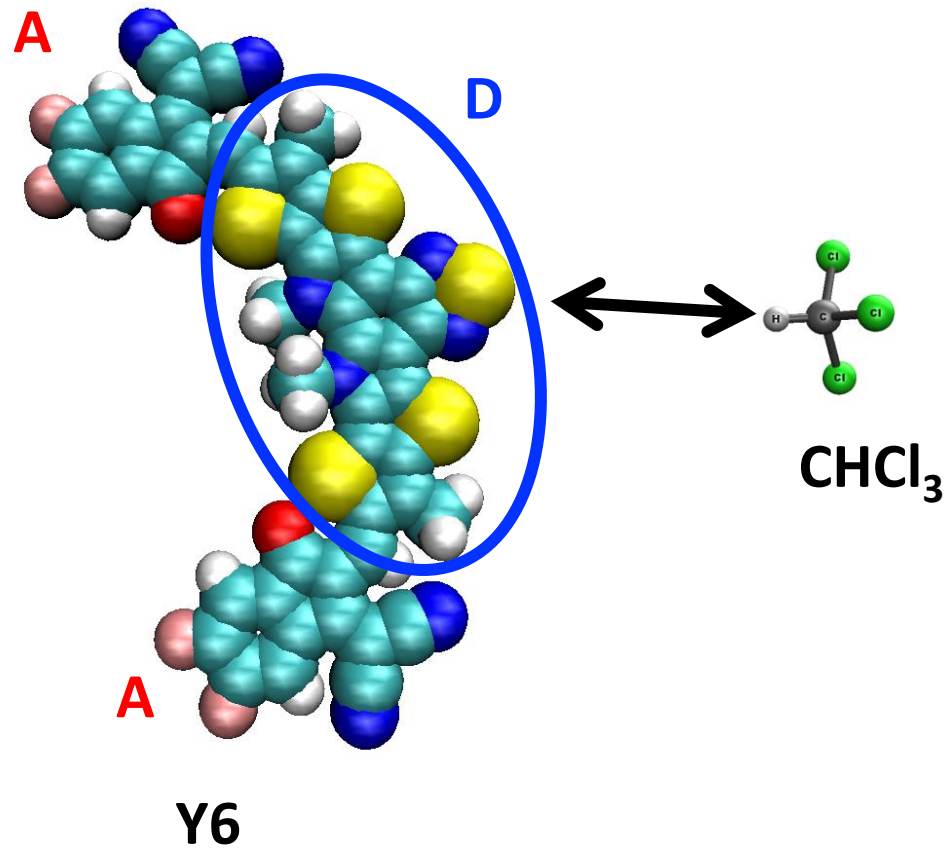
- CT dynamics

$$\Delta Q(t) = \sum_{j \neq g}^F \Delta Q_j \sigma_{jj}(t)$$



All-atom information from directly simulation

- Nuclear dynamics: solvent H-atom RDF near donor surface

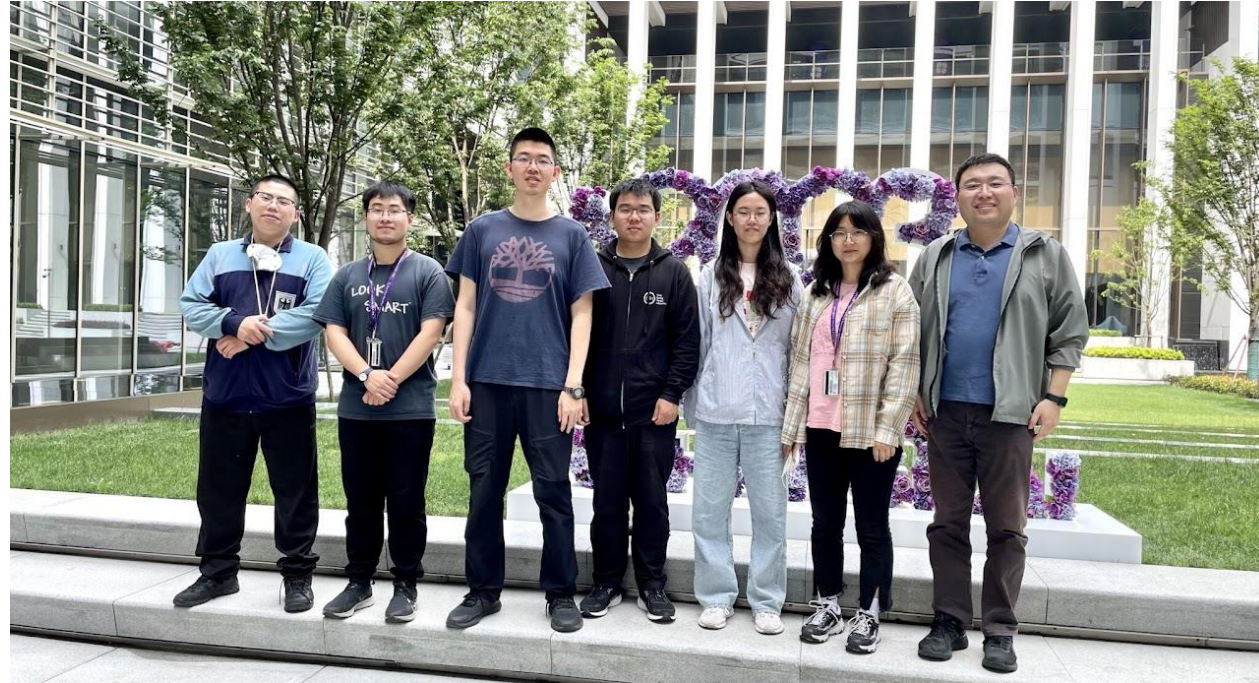


Take-Home Message

- All-atom nonadiabatic semiclassical mapping dynamics can basically reproduce key features of experimental linear absorption spectra for Y6 system.
- Nonperturbative approach provide the atomistic details for all photoinduced charge transfer process.

Acknowledgement

- Greatly thank Professor Xiang Sun, for his exceptional guidance, support, and inspiration throughout my doctoral study.
- Thank my colleagues: Dr. Zhubin Hu, Mr. Dominicus Brian, Dr. Ningyi Lyu, Mr. Hao Zeng, and Ms. Zailing Song.
- Thank Professor Xing Gao for helpful discussion
- Thank our funding agencies: NSFC, SHMEC, and NYUSH



Sun Group, 23 Summer

